

Equity Volatility Strategy

Levered ETF Gamma, RV, Dispersion and FOMC

- For the semi stocks with levered ETF, we combine the dealer's **estimated position from listed gamma** flow with the **effective short gamma arising from their contributions to related levered ETFs**. An intraday momentum strategy on stocks during earnings can help avoid paying premiums at elevated volatility levels ([link](#)).
- We updated the [Dispersion Chartbook \(link\)](#), noting that dispersion levels remain elevated entering earnings season. However, correlation seasonality supports delivery this quarter. We see potential for **sector dispersion within SMH, given high correlations**. IGV and XLV are also candidates to monitor for more attractive entry levels.
- Vol spread between NDX and SMH is at **record highs for a pair with 90% correlation**. We seek a relative value trade between the two, taking advantage of the mismatched earnings calendar, as well as a stand-alone NDX vega trade to anticipate a potential 1998/1999 replay.
 - Buy Aug-26 Call on NDX funded by SMH for credit, Delta-hedged
 - Buy Dec-27 105% UpVar on NDX at 21.9%
- SPX skew and term structure remain elevated, while **record gamma supply from overwriting** continues to suppress near-term realized volatility. We favor using forward-starting downside structures to fund indices **more vulnerable to elevated oil prices and rates**.
 - Buy DAX/SPX Dec26/Jun27 70/110 CCVS at -2.1v (synchronous)
 - Buy RTY/SPX Dec26/Jun27 50/110 CCVS at +2.4v
- We updated the **FOMC straddle premium** cross assets highlighting the muted premium in FX. While FOMC moves during earnings season tend to be less pronounced, we favor a cost-effective macro hedge to cover potential event risk, **including the September meeting**.
 - Buy Sep-26 VIX 25-30 Call Spread Funded by Sell 16.5 Put for 0.13 Net Credit (Ref 19.77)

Global Quantitative and Derivatives Strategy

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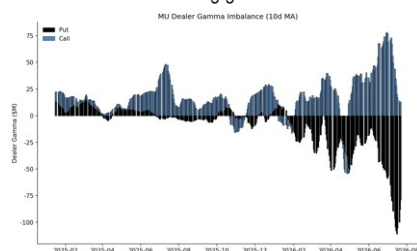
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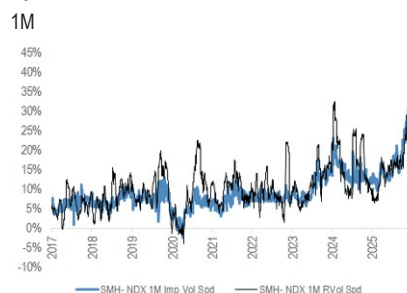
Figure 1: Dealer Gets Shorter Gamma on Puts from Listed Option Only on MU

Positive means dealer long gamma



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 2: SMH Implied and Realized Vol Spread vs NDX



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

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See page 12 for analyst certification and important disclosures.

Options vs ETF Short Gamma in Semis

Our deep dive into **intraday momentum strategies** ([link](#)) showed the recent PnL has been amplified on the names with **large net short gamma effect from levered ETFs rebalancing**, with better performance in **high volatility names on high volatility days**.

Earnings season is generally a good PnL period for stocks' intraday momentum, and we take a closer look at the semi cluster for the **gamma position from listed options**, as well as the effective **short gamma from ETF rebal**. We included **US single names with levered ETF on themselves** as well as their contribution to the levered tech index ETF from the **JPM ETF handbook** ([link](#)).

Figure 3: Semi Stocks Listed Gamma (10D MA) and Levered ETF Effective Gamma vs 1M ADV

From Dealer's Perspective

Stock	Mkt Cap (\$mm)	ETF Gamma (\$)	Listed Gamma (\$)	ETF Gamma/ADV	Listed Gamma/ADV	Net Gamma/ADV
NVDA	5,067	- 712,735,729	112,225,674	-2.61%	0.41%	-2.20%
MU	1,124	- 436,341,492	76,293,012	-0.86%	-0.15%	-1.01%
AVGO	1,877	- 326,177,011	10,248,800	-3.67%	-0.12%	-3.79%
AMD	902	- 238,803,305	3,749,835	-1.59%	-0.03%	-1.62%
INTC	518	- 157,624,972	17,346,775	-1.30%	0.14%	-1.16%
AMAT	450	- 148,740,629	426,281	-2.15%	0.01%	-2.14%
LRCX	402	- 128,551,010	19,807,229	-2.82%	-0.44%	-3.26%
KLAC	283	- 119,169,028	145,512	-3.60%	0.00%	-3.60%
SNDK	242	- 112,317,044	67,841,106	-0.47%	-0.28%	-0.75%
TXN	259	- 106,194,411	706,572	-3.94%	0.03%	-3.91%
MRVL	187	- 102,215,093	4,386,001	-1.45%	0.06%	-1.39%
ASML	699	- 97,657,001	1,891,110	-2.57%	0.05%	-2.52%
ADI	185	- 92,136,349	748,950	-5.12%	-0.04%	-5.16%
TSM	2,176	- 85,335,094	64,162,272	-1.31%	-0.99%	-2.30%
QCOM	183	- 77,883,626	5,911,305	-2.43%	0.18%	-2.25%
NXPI	70	- 56,221,895	2,037,840	-5.16%	-0.19%	-5.35%
MPWR	69	- 54,481,080	59,130	-3.73%	0.00%	-3.72%
TER	59	- 43,784,773	161,466	-2.57%	0.01%	-2.56%
ALAB	57	- 43,759,679	-	-2.34%	0.00%	-2.34%
COHR	63	- 41,586,283	3,611,860	-2.64%	-0.23%	-2.87%
ARM	302	- 41,224,709	-	-2.05%	0.00%	-2.05%
MCHP	45	- 36,393,472	6,236,601	-3.57%	-0.61%	-4.19%
WDC	209	- 30,044,838	4,218,362	-0.59%	-0.08%	-0.67%
STX	209	- 29,392,526	2,638,362	-0.66%	-0.06%	-0.72%
CRDO	43	- 28,752,061	-	-1.41%	0.00%	-1.41%
ON	35	- 24,274,411	2,830,201	-2.03%	-0.24%	-2.26%
GFS	32	- 21,241,412	72,342	-9.24%	-0.03%	-9.28%
MTSI	22	- 14,019,701	-	-2.37%	0.00%	-2.37%
ENTG	21	- 13,894,999	7,619	-2.86%	0.00%	-2.87%
CDNS	93	- 13,863,627	380,480	-1.78%	0.05%	-1.74%
SNPS	72	- 11,005,377	126,693	-1.30%	-0.01%	-1.31%
RMBS	11	- 7,437,387	-	-1.79%	0.00%	-1.79%

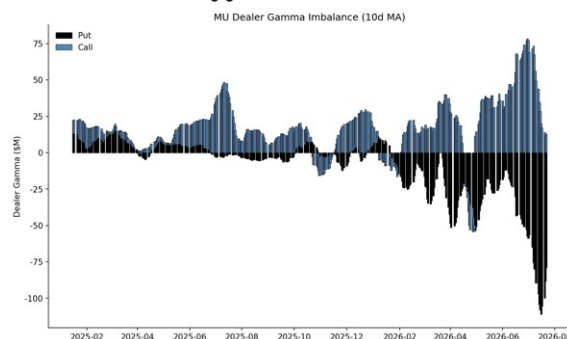
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

We also want to flag that for the biggest names such as MU, we have seen a **pickup in gamma demand on the put, funded by selling calls** from a month ago (Figure 4). The levered ETF flow would **make the situation worse** until more supply picks up. The reduction on gamma supply has been visible across most names in general.

We flagged the increase in intraday momentum PnL in MU compared to NVDA where the result was much worse due to a balancing supply of gamma to dealers from overwriters ([link](#)).

Figure 4: Dealer Gets Shorter Gamma on Puts from Listed Option Only on MU

Positive means dealer long gamma

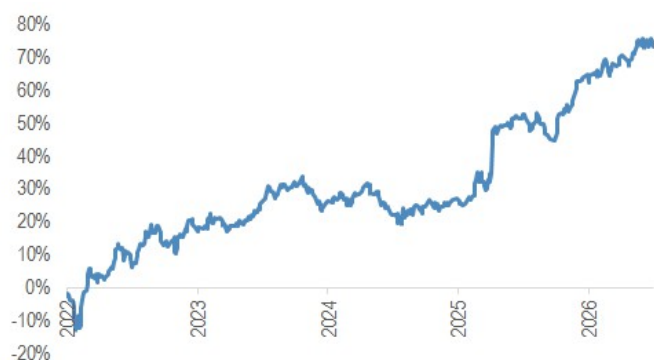


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

We see the same effect for names with **large Net Short Gamma as a % ADV** such as GFS and NXPI, where intraday momentum PnL started to pick up this year.

Figure 5: Intraday Momentum Base Strategy on NXPI

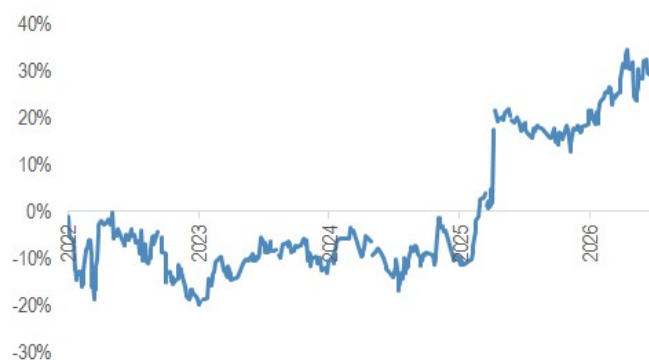
*No Cost



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 6: Intraday Momentum Base Strategy on GFS

*No Cost

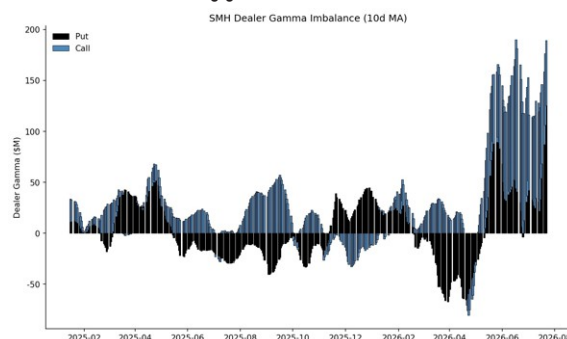


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

That said, we see a moderate **opposite gamma flow from the index level (SMH)**, although more from the call side, one could fund the single stock theta bill with index as a dispersion position into earnings as we discuss next.

Figure 7: Estimation of Dealer Gamma Inventory on SMH

Positive means dealer long gamma

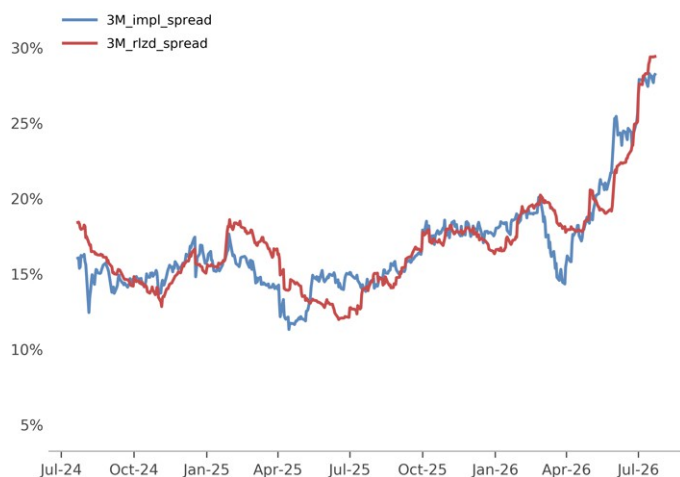


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Dispersion and RV Opportunities

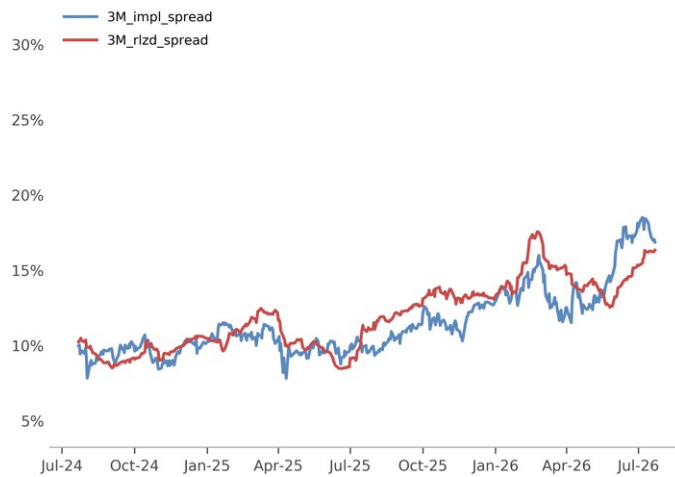
The lack of gamma supply on these names naturally keeps their volatility elevated, both realized and implied. The latest [Dispersion Chartbook update \(link\)](#) shows levels **remain high with elevated expectations** for stock moves into earnings.

Figure 8: SPX T75 Vol Spread Dispersion



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 9: SX5E Vol Spread Dispersion



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Sector Dispersion Trades

However, if we zoom into sector levels, **Semis sector SMH** dispersion looks relatively attractive due to the recent increase in correlation within the sector (Figure 8), as well as the relatively low volatility in NVDA.

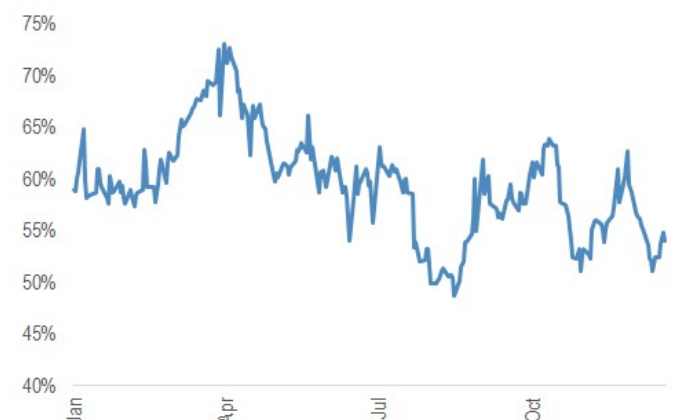
Figure 10: SMH Implied Correlation on the Highs



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 11: Average Realized Correlation for SMH Constituents Trough in Summer and Year-End

From 2020 - 2026



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

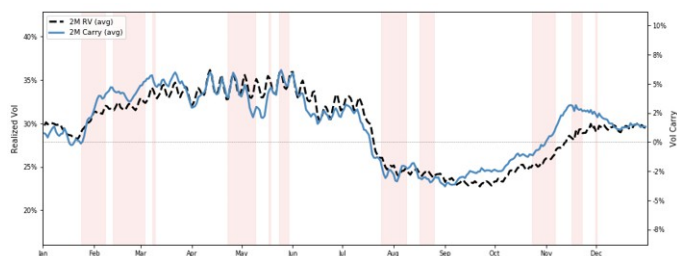
The sector has already gone through a correction from the extreme sentiment ([link](#), [link](#)). At elevated correlation levels, the risk premium is more on the index leg.

We discussed previously the **seasonality of dispersion** during earnings, and July-August earnings are usually a good period with low realized correlation (Figure 11). Investigating further within the tech names, we notice better carry during the Q2 earnings **coming more from the sector index leg**.

Figure 12 shows the realized volatility as well as the proxy carry (Realized volatility minus ATM implied 2M ago) for SMH since 2017, where the trough is during the August period. The same measure for proxy carry in dispersion shows a similar pattern, and we would receive more carry if sizing closer to Theta neutral (Figure 13).

Figure 12: SMH Index Realized Volatility and Proxy Long Vol Carry Seasonality

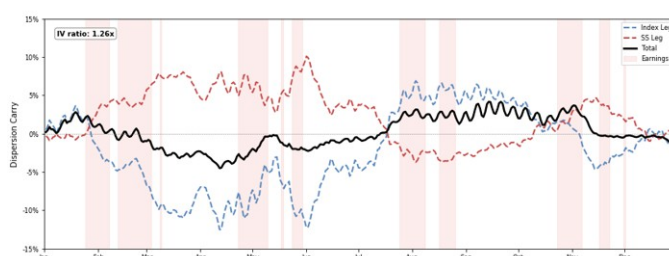
2M, From 2017, Excluding 2020, 2024



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 13: SMH Dispersion Proxy Carry Seasonality, Scaled for Theta Neutral Today

2M, Average of 2021, 2022, 2023, 2025, 2026



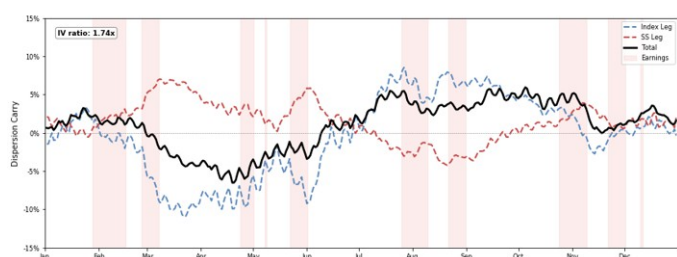
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

With this backdrop, we like owning **SMH dispersion**, preferably **theta weighted**, to cover up to the **September expiry**.

We observe a similar chart for sectors with names that tend to deliver large earnings moves, in particular **IGV (Software)** and **XLV (Healthcare)**. While waiting for a better entry level against the stocks, one can position for volatility spread of the sector index against the major indices ([link](#)).

Figure 14: IGV Dispersion Proxy Carry Seasonality, Scaled for Theta Neutral Today

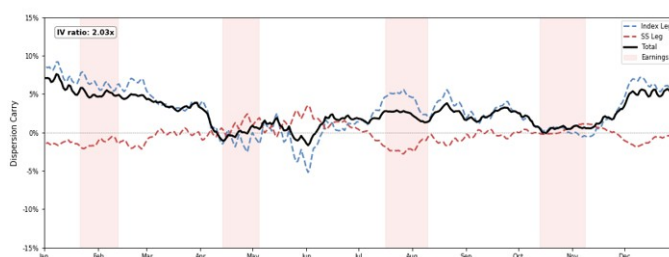
2M, Average of 2021, 2022, 2023, 2025, 2026



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 15: XLV Dispersion Proxy Carry Seasonality, Scaled for Theta Neutral Today

2M, Average of 2021, 2022, 2023, 2025, 2026



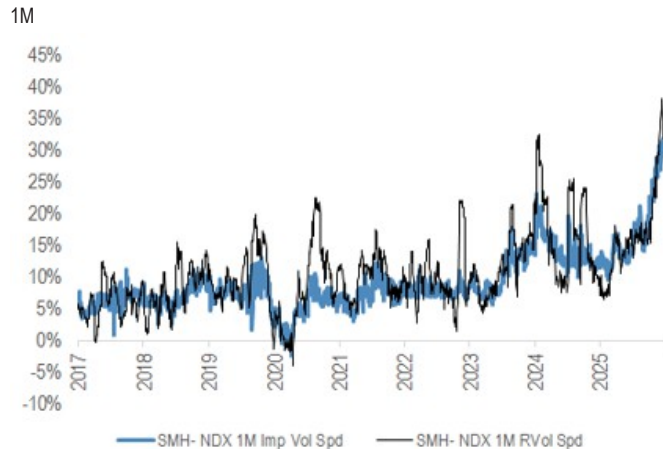
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

As the correlation is at the lows for the broader index, one could place a macro hedge with **September zero cost VIX options** to cover potential correlation spikes (see FOMC section of the report).

RV Trades in NDX

Apart from owning stock gamma against shorting SMH index, another solution is a relative value trade in the **short tenor of SMH against NDX**. Despite having a **correlation of more than 90%** between the two, the volatility spread is now at a record high from the recent semi moves (Figure 16).

Figure 16: SMH Implied and Realized Vol Spread vs NDX



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 17: SMH Implied Vol Spread vs NDX

	1M	2M	3M	6M	1Y	2Y	3Y	4Y
70%F	22.8%	22.4%	21.8%	20.4%	19.5%	20.4%	21.1%	21.1%
80%F	24.8%	24.6%	23.7%	22.2%	20.9%	21.3%	21.9%	22.1%
90%F	27.2%	26.6%	25.7%	24.1%	22.2%	22.1%	22.6%	22.9%
100%F	30.2%	29.3%	28.2%	26.0%	23.6%	23.0%	23.3%	23.7%
110%F	33.3%	31.4%	29.9%	27.3%	24.7%	23.6%	23.9%	24.3%
120%F	26.8%	30.1%	29.5%	27.7%	25.0%	24.0%	24.3%	24.8%
130%F	21.6%	25.6%	26.9%	26.4%	24.7%	24.2%	24.6%	25.2%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

One opportunity to play the spread is in the **upside in very front months**, where the implied vol spread is the highest (Figure 17). We also notice a detailed difference in the constituents between NDX and SMH, that most of the **big weights in SMH either have already reported or will report after August expiry**. In contrast, the NDX will have more heavy weight names prior August expiry, as well as SPCX, which is due to report on Aug 4

Figure 18: Expected Earnings Date for SMH Stocks

Ticker	Weight in SMH	Weight in NDX	Earnings Date	Before Aug Expiry
NVDA	20.4%	8.3%	27-Aug	No
TSM	9.3%	0.0%	16-Jul	No (Reported)
AVGO	6.1%	3.1%	04-Sep	No
AMD	5.9%	4.0%	05-Aug	Yes
MU	5.4%	5.1%	23-Sep	No
ASML	5.1%	0.7%	15-Jul	No (Reported)
AMAT	5.0%	2.0%	14-Aug	Yes
TXN	4.6%	1.2%	23-Jul	No (Reported)
KLAC	4.5%	1.3%	29-Jul	Yes
LRCX	4.4%	1.8%	29-Jul	Yes
INTC	4.4%	2.3%	24-Jul	No (Reported)
ADI	4.3%	0.8%	20-Aug	Yes
QCOM	4.0%	0.8%	30-Jul	Yes
MRVL	3.7%	0.8%	28-Aug	No
CDNS	2.2%	0.4%	28-Jul	Yes
SNPS	1.8%	0.3%	27-Aug	No
MPWR	1.4%	0.3%	31-Jul	Yes
TER	1.3%	0.3%	29-Jul	Yes
NXPI	1.3%	0.3%	28-Jul	Yes
STM	1.2%	0.0%	23-Jul	No (Reported)
ARM	1.1%	0.5%	29-Jul	Yes
ALAB	0.9%	0.3%	05-Aug	Yes
MCHP	0.9%	0.2%	07-Aug	Yes
ON	0.6%	0.0%	04-Aug	Yes
SWKS	0.2%	0.0%	29-Jul	Yes

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Before the August expiry, SMH could be range bound while waiting for the clearing events after expiry from at least NVDA, AVGO, and MU, which makes almost **32% of the index**, and the five stocks already reported forming **almost 25%** could also be less

volatile. Meanwhile NDX could deliver more volatility compared to the current entry level of its implied volatility spread to SMH.

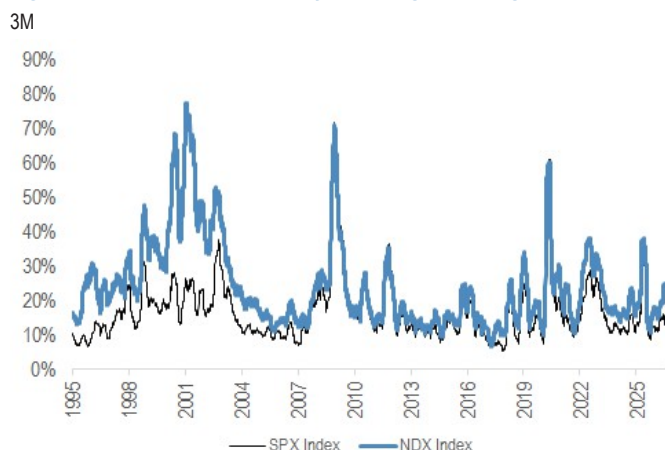
Therefore, we favor a gamma spread between the two as delta-hedge options. Indicatively,

Buy NDX Aug-26 105% Call Sell SMH Aug-26 110% Call for 1.5% Premium Credit, Delta-hedging both legs

The short leg on **SMH** should be **delta hedged intraday** as discussed previously ([link](#)).

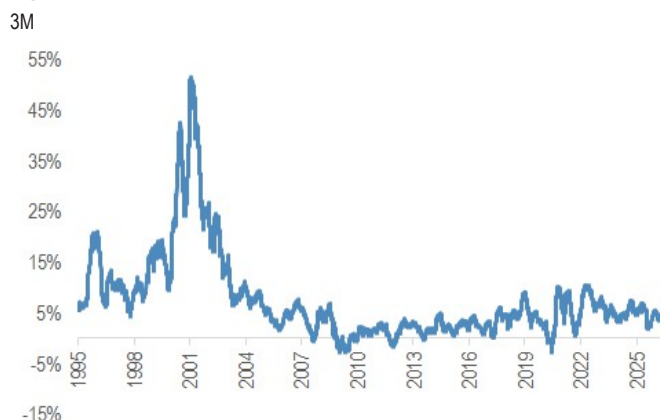
We also discussed previously the resemblance of the current market to 1998/1999, where **high volatility in single stocks, especially the tech cluster, was sustained for a long period of time** ([link](#), [link](#)). While correlation was very low within the broader SPX index, bringing index vol to a muted level, **the tech cluster had a higher correlation**, making NDX realized volatility much higher.

Figure 19: NDX Realized Volatility Much Higher During Dot-Com



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 20: NDX- SPX Realized Vol Spread



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

With NDX consolidating near the June lows, 8% off the peak, one can use an UpVar on NDX to **participate in the potential rebound without paying premium**. If the market sells off further from here until expiry, the option would have zero vega hence **no PnL**. Should the market rebound later on, one can collect the variance above the barrier to position for an **upside further overshoot followed by correction at a near 10 vol discount from the varswap**, while the mark-to-market converges toward a varswap in the meantime ([link](#)).

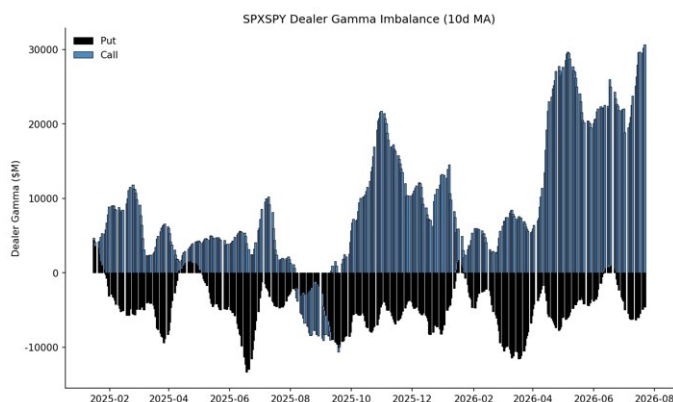
To maximize the probability of being in the range, and the likelihood of positive mark-to-market, a longer tenor is preferred. We favor a near one-and-a-half-year maturity, indicatively,

Buy Dec-27 105% UpVar on NDX at 21.9%

Index RV with SPX

The increased activity from overwriting has driven **gamma supply in the SPX to near record highs** (Figure 21). Subsequently, SPX downside skew is very elevated (Figure 22), and term structure remains steep from the low volatility in the front month. Against this backdrop, SPX could potentially be used as a funding leg for owning other index vega, with **forward-starting downside** most attractive.

Figure 21: Dealer's Gamma Inventory on the Highs from Listed Observation



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 22: SPX Skew Back to the Highs



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Europe, situated between two major conflict zones—from Ukraine to the Middle East—and facing additional headwinds from the latest hawkish ECB update, could be particularly vulnerable to any potential growth slowdowns. DAX is one of the worst performers year to date and, notably, **lacks any Oil companies to help diversify price risk**. We favor the trade at a **2 vol discount to the spot starting** version, indicatively,

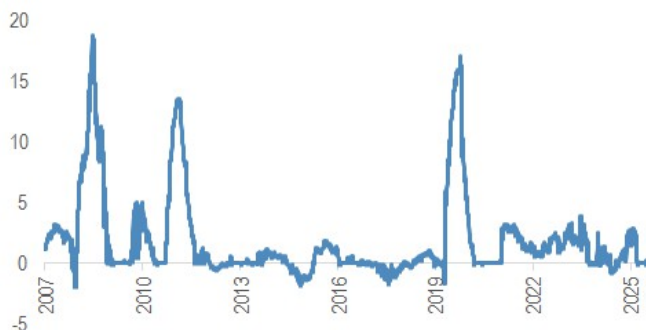
Buy DAX/SPX Dec26Jun27 70/110 CCVS at -2.1v, Barrier Observation on DAX (Quanto USD London 430pm synchronous)

Figure 23: Back-test for DAX/SPX CCVS at Current Entry



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 24: Back-test for RTY/SPX CCVS at Current Entry



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

A similar trade can be implemented in RTY, where elevated rates could be more hurtful due to the sensitivity of the cluster. Indicatively,

Buy RTY/SPX Dec26/Jun27 50/110 CCVS at +2.4v. Barrier Observation on RTY

FOMC

The option premium covering FOMC date next week indicates **FX has the lowest expectations** on the size of the move compared to historical averages. Within equities, **SX5E, RTY, and SPX** are the cheapest to own compared to their historical averages.

Figure 25: Cross Asset Straddle Premium vs Avg Historical Intraday Moves in FOMC Dates

	Straddle Cost	Straddle -	Historical		Historical Intraday H-L								Historical Close to Close							
	Jul 31	Hist Avg	Intraday	Close-Close	Jun-26	Apr-26	Mar-26	Jan-26	Dec-25	Oct-25	Sep-25	Jul-25	Jun-26	Apr-26	Mar-26	Jan-26	Dec-25	Oct-25	Sep-25	Jul-25
SPX	1.86%	0.83%	1.03%	0.36%	1.75%	0.53%	1.24%	0.56%	1.11%	0.99%	1.11%	0.94%	0.24%	0.40%	-1.36%	-0.01%	0.67%	0.00%	-0.10%	-0.12%
NDX	3.04%	1.86%	1.18%	0.49%	2.04%	0.76%	1.40%	0.73%	1.29%	1.06%	1.20%	0.94%	0.22%	0.71%	-1.43%	0.32%	0.42%	0.41%	-0.21%	0.16%
RTY	2.23%	0.31%	1.92%	0.72%	2.28%	1.30%	1.31%	1.43%	2.26%	2.39%	2.38%	2.03%	0.24%	0.52%	-1.64%	-0.49%	1.32%	-0.87%	0.18%	-0.47%
SX5E	1.76%	0.77%	0.98%	0.40%	0.74%	0.93%	2.06%	1.52%	0.53%	0.74%	0.61%	0.75%	0.74%	0.34%	-0.56%	-1.02%	-0.18%	0.03%	-0.05%	0.26%
EEM	3.78%	2.47%	1.31%	0.77%	2.43%	1.09%	1.84%	0.83%	1.21%	1.01%	1.29%	0.79%	0.22%	0.40%	-2.03%	0.68%	1.12%	0.57%	0.43%	-0.73%
FXI	3.30%	2.09%	1.20%	0.90%	2.26%	0.94%	1.41%	0.90%	0.89%	1.22%	0.96%	1.03%	0.16%	0.25%	-1.87%	1.40%	0.70%	0.27%	1.39%	-1.15%
EURUSD	0.68%	-0.29%	0.97%	0.57%	1.21%	0.51%	0.91%	1.24%	0.67%	0.76%	0.94%	1.49%	0.20%	0.14%	-0.76%	-0.72%	0.58%	-0.43%	-0.46%	-1.23%
JPYUSD	0.79%	-0.09%	0.88%	0.43%	0.42%	0.59%	0.83%	1.20%	0.74%	1.00%	1.07%	1.17%	0.10%	0.03%	-0.54%	-0.79%	0.56%	-0.40%	-0.35%	-0.70%
CHFUSD	0.81%	-0.29%	1.10%	0.67%	1.33%	0.52%	1.14%	1.50%	0.92%	1.18%	0.83%	1.38%	0.22%	0.16%	-1.05%	-0.92%	0.81%	-0.81%	-0.35%	-1.08%
GBPUSD	0.73%	-0.17%	0.90%	0.45%	1.31%	0.52%	0.92%	0.71%	0.70%	1.05%	0.78%	1.18%	0.23%	0.13%	-0.74%	-0.30%	0.65%	-0.59%	-0.15%	-0.85%
GLD	2.52%	0.52%	2.00%	1.44%	3.86%	1.21%	1.23%	3.07%	1.36%	2.39%	1.68%	1.21%	0.39%	0.78%	-3.16%	3.88%	0.43%	-0.38%	-0.77%	-1.73%
USO	7.67%	4.96%	2.71%	1.37%	5.14%	3.90%	4.56%	1.47%	2.25%	1.25%	1.36%	1.75%	0.09%	3.21%	2.38%	1.27%	0.97%	0.57%	-0.93%	1.55%
HYG	0.78%	0.39%	0.39%	0.23%	0.45%	0.45%	0.45%	0.18%	0.45%	0.41%	0.41%	0.34%	0.03%	0.19%	-0.51%	-0.16%	0.34%	-0.31%	-0.10%	-0.21%
LQD	0.78%	0.26%	0.52%	0.31%	0.55%	0.43%	0.56%	0.25%	0.57%	0.62%	0.73%	0.43%	0.06%	0.15%	-0.54%	-0.14%	0.46%	-0.57%	-0.18%	-0.37%
TLT	1.29%	0.56%	0.73%	0.45%	0.77%	0.51%	0.63%	0.47%	0.76%	0.95%	1.29%	0.46%	0.47%	0.18%	-0.57%	-0.23%	0.39%	-1.01%	-0.25%	-0.52%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

That said, we can't ignore the timing of the Fed meeting being **in the middle of the busiest reporting week**. Historically, the realized move on the index level for equities was **lower on dates of FOMC falls in earnings periods compared to outside**, as the market waits for clearing events from stock reports.

Figure 26: Average Intraday High-Low for Major Equity Index on FOMC Dates Falling in Earnings Periods vs Non-Earnings Periods

2017-2026, excluding 2020, (High-Low)/Previous Close

Ticker	Earnings	NonEarnings
SPX Index	1.31%	1.42%
NDX Index	1.68%	1.81%
RTY Index	1.80%	2.03%
SX5E Index	1.02%	1.10%
EEM US Equity	1.30%	1.52%
FXI US Equity	1.47%	1.63%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

With VIX at current muted levels, we can place a macro hedge with **September VIX options**. This tenor should have a floor as it will reference to SPX implied volatility in **October**, which not only covers the **September FOMC** but is also close to the **November midterm elections**.

Buy Sep-26 VIX 25-30 Call Spread Funded by Sell 16.5 Put for 0.13 Net Credit (Ref 19.77)

Appendix – Recent Publications

Equity Volatility Strategy

[Intraday Momentum Deep Dive \(Part I\)](#)

[Dispersion, Mag7 Volatility Opportunity And Inverted Skew](#)

[Dispersion At The Extremes - What Happens After a 100% 1M Return](#)

[Gamma Supply and Earnings Dispersion Update](#)

[Have Parameters Normalized Too Much?](#)

[What To Long, Short And When \(Part II\)](#)

[“Too Big to Move”? Dispersion Update and RV Opportunities](#)

[Lesson From The 2022 Playbook](#)

[Vol RV opportunities amid the software correction](#)

[What to Long, Short, and When \(Part I\)](#)

[Why index vol is so low and MXWO dispersion](#)

[Term-structure inversion, NVDA, Gamma dispersion and VIX](#)

[Thinking Outside the Box for Tail Trading](#)

[Thinking Outside the Box for Tail Trading - Part 2](#)

[Correlation and Volatility at the Lows: What to do](#)

[Take some chips off the table, replace with leveraged option](#)

[Trading US macro events with short-dated options](#)

[The case for long dated Upvar](#)

Comprehensive Reports on Dispersion

Sep 2023: [A Practical Dispersion Trading Handbook](#)

Sep 2024: [Systematic Dispersion Update and Trade Ideas](#)

Apr 2025: [Dispersion Update, Index Technique, Stock Leg and What Now](#)

Oct 2025: [Dispersion Update, AI vs Dot-Com and Earnings](#)

Cross Asset Volatility Strategy

[What to Long, Short, and When - \(Commodity Part I\)](#)

[Trend, Oil and Gold Derivatives and ML Model Update](#)

[2022 Deep Dive, Oil and Gold Update, FOMC and ML Model Update](#)

[Oil Derivatives and “Cheaper To Deliver” Macro Hedges](#)

[Gold Through the Lens of Derivatives: Positioning and Trade](#)

[FOMC Implied Moves, AI Credit Hedge and ML Model Output Update](#)

[Record Short Gamma in Gold ETF and Machine Learning Based Trade Recommendations](#)

Risks of Common Option Strategies

Risks to Strategies: Not all option strategies are suitable for investors; certain strategies may expose investors to significant potential losses. We have summarized the risks of selected derivative strategies. For additional risk information, please call your sales representative for a copy of “Characteristics and Risks of Standardized Options.” We advise investors to consult their tax advisors and legal counsel about the tax implications of these strategies. Please also refer to option risk disclosure documents.

Put Sale: Investors who sell put options will own the underlying asset if the asset’s price falls below the strike price of the put option. Investors, therefore, will be exposed to any decline in the underlying asset’s price below the strike potentially to zero, and they will not participate in any price appreciation in the underlying asset if the option expires unexercised.

Call Sale: Investors who sell uncovered call options have exposure on the upside that is theoretically unlimited.

Call Overwrite or Buywrite: Investors who sell call options against a long position in the underlying asset give up any appreciation in the underlying asset’s price above the strike price of the call option, and they remain exposed to the downside of the underlying asset in the return for the receipt of the option premium.

Booster : In a sell-off, the maximum realized downside potential of a double-up booster is the net premium paid. In a rally, option losses are potentially unlimited as the investor is net short a call. When overlaid onto a long position in the underlying asset, upside losses are capped (as for a covered call), but downside losses are not.

Collar: Locks in the amount that can be realized at maturity to a range defined by the put and call strike. If the collar is not costless, investors risk losing 100% of the premium paid. Since investors are selling a call option, they give up any price appreciation in the underlying asset above the strike price of the call option.

Call Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is below the strike price of the call option.

Put Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is above the strike price of the put option.

Straddle or Strangle: The seller of a straddle or strangle is exposed to increases in the underlying asset’s price above the call strike and declines in the underlying asset’s price below the put strike. Since exposure on the upside is theoretically unlimited, investors who also own the underlying asset would have limited losses should the underlying asset rally. Covered writers are exposed to declines in the underlying asset position as well as any additional exposure should the underlying asset decline below the strike price of the put option. Having sold a covered call option, the investor gives up all appreciation in the underlying asset above the strike price of the call option.

Put Spread: The buyer of a put spread risks losing 100% of the premium paid. The buyer of higher-ratio put spread has unlimited downside below the lower strike (down to zero), dependent on the number of lower-struck puts sold. The maximum gain is limited to the spread between the two put strikes, when the underlying is at the lower strike. Investors who own the underlying asset will have downside protection between the higher-strike put and the lower-strike put. However, should the underlying asset’s price fall below the strike price of the lower-strike put, investors regain exposure to the underlying asset, and this exposure is multiplied by the number of puts sold.

Call Spread: The buyer risks losing 100% of the premium paid. The gain is limited to the spread between the two strike prices. The seller of a call spread risks losing an amount equal to the spread between the two call strikes less the net premium received. By selling a covered call spread, the investor remains exposed to the downside of the underlying asset and gives up the spread between the two call strikes should the underlying asset rally.

Butterfly Spread: A butterfly spread consists of two spreads established simultaneously – one a bull spread and the other a bear spread. The resulting position is neutral, that is, the investor will profit if the underlying is stable. Butterfly spreads are established at a net debit. The maximum profit will occur at the middle strike price; the maximum loss is the net debit.

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